

The adaptive plain-moment estimate: supplementary proof

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Abstract

We extend the parameter range of the plain fourth-moment estimate in OpenAI's September 2026 preprint from $\kappa \geq 3/4$ to $\kappa \geq 13/18$, assuming the corresponding global zero bound for finite-order Hecke characters. We verify the comparison inequalities, the admission of every recursive child, and the order of the mesh, height and induction choices. We then give the resulting witness count. The unchanged arithmetic identities and analytic estimates are inputs from the cited, unrefereed preprint. This supplement does not assert an independent proof of those inputs or perform the subsequent zero-free-region continuation.

1 The estimate and its hypotheses

Put $F = \mathbb{Q}(\sqrt{-3})$. We use the primary generators, sextic residue symbols and fixed finite set of excluded primes $\mathcal{S}$ of [1, Sections 4 and 14]. For an integral ideal or its primary generator n , write $q_n = Nn$; for an element k , write $q_k = |k|^2$. Sums over n or l use the primary representatives of ideals prime to $\mathcal{S}$. The character $\chi_n(k)$ is the zero-extended sextic residue symbol. In particular, a displayed sixth power is one on units and is zero on nonunits.

Let β_* be the supremum of $1/2$ and of the real parts of zeros in $1/2 \leq \operatorname{Re} s \leq 1$ of primitive finite-order Hecke L -functions over F . Principal poles are excluded. This is the global family quantity in [1, Section 2, (2.1)].

Fix a finite group Θ of finite-order ray characters supported on $\mathcal{S}$, containing the fixed twists, the characters used to separate reciprocity phases, and $n \mapsto \chi_n(-1)$. At scale $Z \geq 2$, the rows and their characters are

$$0 < q_k \ll Z^m, \quad \psi_k(n) = \tau(n)\chi_n(k), \quad M = m + q.$$

Here τ is fixed within the row sum. The union of the displayed moving prime supports in τ has norm at most Z^q . It is measured *before* cancellation or reduction of exponents modulo six. Every factor retains its original zeros. An additional squarefree mask $\mathfrak{R}$, of norm at most Z^{B_0} for fixed B_0 , may depend on frozen parameters but is fixed within a row sum and is common to both plain sums and every prime factor.

For smooth profiles supported on fixed annuli, put

$$S_\psi(n; W) = Z^{-n/2} \sum_l \psi(l) \mathbf{1}_{(l, \mathfrak{R})=1} W(q_l/Z^n), \tag{1}$$

$$Q_{\psi, i} = Z^{-z_i/2} \sum_{p \text{ prime}} \psi(p) \mathbf{1}_{(p, \mathfrak{R})=1} \nu_i(p) W_i(q_p/Z^{z_i}), \quad Q_\psi = \prod_i Q_{\psi, i}, \quad z = \sum_i z_i. \tag{2}$$

Each ν_i is a fixed finite linear combination of members of Θ , independent of the row and all other prime choices. The underlying prime windows are disjoint before imposing any zeros or masks. A factor of length zero has bounded scale and is estimated by absolute counting. Let $\mathcal{R}_0$ contain the rows with nonprincipal primitive inducing character. If $z > 0$, let $\mathcal{R}_z$ contain the rows whose primitive inducing character is not induced by a member of Θ . Membership refers to the inducing character, without removing the presentation's zeros.

Theorem 1.1 (Extended plain moment). *Retain the arithmetic and profile hypotheses of [1, Lemma 18.1]. Suppose*

$$\frac{13}{18} \leq \kappa \leq \frac{3}{4}, \quad \beta_* \leq s_\kappa := \frac{1 + \kappa}{2}.$$

For every $\varepsilon > 0$ and prescribed bounded ranges of nonnegative lengths, there is a mesh $\eta > 0$, chosen before the fixed number of prime factors, such that, whenever $z_i \leq \eta$,

$$\sum_{k \in \mathcal{R}_z} |S_{\psi_k}(n_1; W_1) S_{\psi_k}(n_2; W_2) Q_{\psi_k}|^2 \ll Z^{M+\varepsilon}. \quad (3)$$

For $z = 0$, the lengths n_1, n_2 are unrestricted in their prescribed bounded ranges, and no hypothesis on β_ is required. For $z > 0$, assume*

$$n_1 + n_2 + 6\kappa z \leq M. \quad (4)$$

The estimate retains the finite smooth-seminorm and polynomial height bounds of the original lemma. Their orders and constants may depend on the fixed data and number of prime factors, but are uniform in rows, moving labels, masks and the numerical heights. The mesh is uniform for $\kappa \in [13/18, 3/4]$.

The global hypothesis is essential: the two Poisson transforms create new finite-order characters, which need not have the same rightmost zero as the original row. We prove the extension by checking every use of the lower bound on κ in [1, Section 18]. Its exact reflection, finite correlations, common-coefficient identity and masked lattice cancellation are imported with all their original hypotheses. The argument below changes their numerical admissions; it does not replace their coefficients by arbitrary bounded weights.

2 Prime bounds and comparison lengths

For a row inducing outside Θ , every character obtained by multiplying it by a component of ν_i is nonprincipal. Otherwise the inducing row character would belong to Θ . This remains true for each recursive child. The global hypothesis and [1, Lemma 4.9] therefore permit the prime contour to move to $\text{Re } s = s_\kappa + e$, for fixed $e > 0$, uniformly in the conductor and height. The von Mangoldt formulation, removal of higher prime powers and central normalization give

$$|Q_\psi|^2 \ll Z^{\kappa z + \varepsilon_1} p_J(\mathbf{W})^b (1 + |\boldsymbol{\omega}|)^h. \quad (5)$$

Here p_J is a finite smooth seminorm, and $\boldsymbol{\omega}$ records the independent normalized twists of the profiles. The contour displacement costs $2ez$, so it is controlled by the bounded *total* length, not by the number of factors. The orders J, b, h are fixed after the profiles and requested derivatives. They do not add a power of Z .

Extra masks are removed by the exact identities

$$S_{k,\mathfrak{R}}(X) = \sum_{d|\mathfrak{R}} \mu(d) \psi_k^0(d) q_d^{-1/2} T_k(X/q_d),$$

$$Q_{k,i,\mathfrak{R}} = Q_{k,i}^0 - P_i^{-1/2} \sum_{p|\mathfrak{R}} \psi_k^0(p) \nu_i(p) W_i(q_p/P_i), \quad P_i = Z^{z_i}.$$

Here ψ_k^0 retains the natural zeros but omits the additional mask, and T_k is its centrally normalized plain sum. Their absolute coefficient masses are Z^{ε_1} by [1, Lemma 4.10 and Section 18.1]. The inducing character is unchanged; each deletion decreases the left side of (4). If no prime factor survives, one uses the unrestricted zero-factor assertion.

Set $A = n_1 + n_2 + z$. At the small-width endpoint, admission gives

$$z \leq \frac{M}{6\kappa} \leq \frac{3M}{13}, \quad \kappa z \leq M/6. \quad (6)$$

The second inequality is the required terminal exponent payment. The slightly larger length range in the first inequality has no analytic consequence.

For the comparison step take $L = M/4$. If $A > 5M/6$, then

$$(6\kappa - 1)z \leq M - A, \quad 6\kappa - 1 \geq 10/3, \quad \text{hence} \quad z < M/20.$$

If one original plain factor has length below L , reflect the other. Otherwise compare the original product with a product having lengths $L, A - z - L$, the same profiles, and the same product of scales; reflect its longer factor when estimating the comparison.

The primitive functional equation, with the redundant deleted Euler factors restored, gives a reflected length at most $M - n + \xi$ for a factor of length n . This is exactly the reflection of [1, Section 18.1]: its conductor times its redundant radical is bounded by Z^M ; retained subunit annuli are rescaled only after the support test, and both the annular tail and scale supremum are paid by finite seminorms and polynomial heights. Consequently the comparison has

$$A_{\text{comp}} \leq 3M/2 - A + 2z + \xi \leq 23M/30 + \xi, \quad (7)$$

$$A_{\text{comp}} + (6\kappa - 1)z \leq 5M/2 - 2A + 2z + \xi \leq 14M/15 + \xi. \quad (8)$$

Both required boundaries have a margin of $M/15$ before ξ . Thus the comparison uses only the uncentered range already proved at that width.

For completeness, suppose an aggregate clipping error e_c changes the inequality to $A \geq 5M/6 - e_c$. Then $z \leq M/20 + 3e_c/10$, and the two bounds become

$$A_{\text{ref}} \leq 23M/30 + \xi + \frac{8}{5}e_c, \quad A_{\text{ref}} + (6\kappa - 1)z \leq 14M/15 + \xi + \frac{13}{5}e_c. \quad (9)$$

The coefficients $8/5$ and $13/5$ include the perturbation of A itself, not only that of z . The zero branch of the reflected positive part has affine bound at most $7M/15 + \xi + 13e_c/10$. The long comparison factor has length at least $8M/15 - 13e_c/10$. If $M \geq \rho$ and $\xi, e_c \leq \rho/100$, the margins in the uncentered boundary, admission boundary, and the long-factor boundary $M/2$ are respectively

$$\frac{61\rho}{1500}, \quad \frac{23\rho}{750}, \quad \frac{61\rho}{3000}.$$

These are positive uniformly throughout the new interval.

3 Recursive children and the finite induction

We now check that the changed admissions suffice for the entire recursive argument. All the arithmetic steps in this section are the exact identities of [1, Sections 18.3–18.7]. Their proofs are independent of the lower bound on κ . The following list specifies what is imported; the accompanying argument checks that it is applicable in the enlarged interval.

Input from [1]	Data retained in the recursive argument
Section 18.1, (18.4)–(18.10)	Exact deletion, inducing character, redundant natural zeros, reflection scales and conductor–support bound.
Section 18.3, (18.17)–(18.19)	Equal products of scales, common mask, one whole-product norm power, and a support box fixed before the live prime labels.
Section 18.4	Full first Poisson identity and complete common-support allocation; localization on the genuine sum before full Möbius inversion.
Section 18.5	One Gauss-row enlargement, all valuation-one, six and seven errors, and an auxiliary prime pool disjoint from live prime windows.
Section 18.6, Lemma 18.2	Full second correlation, independent child products with their original prime coefficients and zeros, and exact normalizers.
Section 18.7, Lemma 18.3	Exceptional-row count and masked equal-product lattice cancellation on the raw signed rectangles.

3.1 The preserved coefficient

When centering is needed, write $X_1X_2 = Y_1Y_2$ and retain the entire difference of the two profile rectangles. After fixed plain powers $\mathbf{b}_1, \mathbf{b}_2$ have been extracted, it is

$$D_{\mathbf{b}}(l_1, l_2) = \prod_{i=1}^2 W_i(q_{\mathbf{b}_i l_i}/X_i) - \prod_{i=1}^2 W_i(q_{\mathbf{b}_i l_i}/Y_i).$$

Within each separated side, the coefficient is the product of this difference, one common character, one common puncture, one power $q_{l_1 l_2}^{it}$, and the original live prime coefficients. The first transform adds a Gauss coefficient on the whole product. The second transform removes it and restores the same prime coefficients. It does not leave a new coefficient in an individual prime factor [1, Lemma 18.2].

At either transform, common support is extracted with its full multiplicities, not merely its gcd. A prime supplied by a live prime factor is frozen if it enters this support, and punctures every remaining factor on that side. The artificial residual noncoprime pairs are removed by the *full* finite Möbius indicator before forming independent products. Quotients may still contain the divisor primes; no unproved coprimality is imposed on them. Both inverse roots and the full Poisson kernel are retained until their joint Fourier separation. The coefficient measure is fixed on the whole-product support box before live labels are chosen, and is common to both rectangles.

Let N be the fixed number of prime factors. Choose H_N to bound the sum of the absolute logarithms of their relative annular endpoints, enlarged for all plain windows, reflection multipliers, fixed arithmetic normalizations, relative dyadic boxes and support enlargements through the bounded depth. For every subset of factors, its aggregate support correction is at most

$$\theta_N = H_N / \log Z.$$

There is one such correction, rather than N separately chosen length errors. The raw frequency tails are first paid on the genuine source. Only afterwards is a common retained dyad extended by zero to its enclosing row ball.

3.2 Width decrease and child admission

Here are the relevant exact length identities. Let c, d be the logarithmic norms of the two first extracted common-support moduli, and let R be the logarithmic norm of their common primes whose net character exponents are nonzero modulo six. Let E be the logarithmic norm of the divisor selected in the Möbius expansion of the complementary common row mask. Put

$$\tilde{q} = q + R + E, \quad K_0 = 2A - c - d + R + E - m.$$

This is the displayed support union and nominal first frequency length of [1, (18.22)]; let K be its retained dyad length. In one Gauss norm, w is the length removed from its column and $w_o \leq w$ is the new moving support thereby created. Put

$$a_0 = A - c - w, \quad J = d - c + (K_0 - K) - 2w + w_o.$$

The row-ball increment is denoted by g , and the prime-density normalizer by $Z^{-\ell}$. Without prime factors, $\ell = 0$ and $g = J_+ + \sigma$. With prime factors, the enlarged main norm has $\ell = \sigma/3$, $g = J_+ + 2\sigma$. Each extracted valuation error instead has $\ell = 0$, $g = J_+ + \sigma$ and $0 \leq w_o \leq w \leq 7\sigma/3$. Errors are not enlarged again. The auxiliary pool has exponent $\sigma/3$, whereas all live lengths are at most $\eta < \sigma/6$; disjointness follows after a threshold depending only on the fixed windows.

In the second complete-support extraction write the two moduli as D_2a, E_2b , and set

$$c_2 = \log_Z q_{D_2}, \quad d_2 = \log_Z q_{E_2}, \quad b_2 = (c_2 + d_2)/2, \quad G_c = (D_2, E_2), \quad g_2 = \log_Z q_{G_c}.$$

Let p_2 be the length of their common radical. Among equal non-six-divisible common valuations, let t_2 count the primes where the divided frequency is a unit. Let V_{id} be the product of the corresponding nonunit primes, and put $V = \log_Z q_{V_{\text{id}}}$. The complete local correlation, including all zero cases, is partitioned on these unit and nonunit conditions before its scalar is bounded. With $\tilde{q}$ the full moving support left by the first transform, the exact child widths are

$$m' = 2a_0 - K - g - g_2 - V, \quad q' = \tilde{q} + w_o + t_2 + V, \quad (10)$$

$$M' = m' + q' = M + J - g - g_2 + t_2 \leq M - \sigma. \quad (11)$$

This is [1, (18.32)]; $g_2 \geq t_2$ and $g - J \geq \sigma$. The finite-correlation normalizers leave the child the allowance

$$M' + \Delta_{\text{child}}, \quad \Delta_{\text{child}} = b_2 - p_2 + w + \mathcal{B}_c + \ell \geq w + \ell, \quad (12)$$

as in [1, (18.33)]. The nonnegative first-extraction bonus is $\mathcal{B}_c = (3c - 5d - R)_+/6$.

The two child characters have the forms

$$\tau_1(n)\rho(n)\chi_n(G_c V_{\text{id}} h'), \quad \tau_1(n)\rho'(n)\chi_n(-G_c V_{\text{id}} h'), \quad \rho, \rho' \in \Theta.$$

Their inducing-character ratio belongs to Θ ; this statement takes no quotient at a zero. Every moving good prime belongs to the declared union in (10). Six-divisible factors remain common punctures. The remaining choices at $\mathcal{S}$ are finite ray choices, and the norm powers only translate Mellin height. Thus all nonexceptional children still lie in the global family to which (5) applies.

For such children, first take the triangle inequality between the two rectangles, and then apply the support test to each plain scale. An empty annulus is discarded; every retained subunit annulus is rescaled to one. The exact clipped normalizers are those of [1, (18.34)–(18.37)]. In particular the reductions may differ between rectangles. Let $A_{\text{act}}, z_{\text{act}}$ be the actual lengths after all allocations, clipping and mask deletion, and M'_{act} their enclosing width. For the two transforms set

$$\delta_{\text{fr},j} = \frac{\xi}{2} + \frac{\log(C_d C_{\text{sec},j})}{\log Z} < \xi \quad (j = 1, 2).$$

Here C_d bounds the relative width of a frequency dyad and $C_{\text{sec},j}$ bounds the ratio of the nominal frequency scales within its fixed outer sector. The eventual threshold ensures the displayed inequalities. Retain $K_0 - K \geq -\delta_{\text{fr},1}$ and $0 \leq M'_{\text{act}} - M' \leq \delta_{\text{fr},2}$. The entire admission defect is

$$F_{\text{act}} = (A_{\text{act}} - M'_{\text{act}} + (6\kappa - 1)z_{\text{act}})_+.$$

The exact ledger [1, (18.38)], including the clipped normalizers, gives

$$F_{\text{act}} \leq 6(w + \ell) + \delta_{\text{fr},1} + 2\theta_N.$$

Remove one prefix of whole prime factors until admission holds, or until none remains. Since every length is at most η , its total removed length d_z satisfies

$$\kappa d_z \leq F_{\text{act}}/6 + \kappa\eta \leq \Delta_{\text{child}} + \delta_{\text{fr},1}/6 + \theta_N/3 + \eta.$$

This uses only $6\kappa - 1 \geq 0$ and $0 < \kappa \leq 1$. There is one mesh overshoot. If all prime factors disappear, the remaining plain lengths use the completed unrestricted zero-factor estimate. After pairing the two children and their exact coefficients, the extra edge cost is

$$\delta_{\text{fr},2} + \delta_{\text{fr},1}/6 + \eta + 7\theta_N/3, \tag{13}$$

the bound of [1, (18.40)]. The already allocated discrete and analytic costs remain separate.

3.3 Exceptional rows

If the inducing character belongs to Θ , keep the raw signed rectangles. Neither clipping nor the nonprincipal prime bound is used on them. After fixing the live prime labels, Lemma 18.2 of [1] supplies one character, one common polynomial-size mask, and one norm power in both plains and both rectangles. The lattice estimate of [1, Lemma 18.3] has main term

$$c_{\vartheta, \mathfrak{R}_*} X^{1+it} I_i(t),$$

whose coefficient is independent of X, t . Its two product main terms therefore agree at equal products of scales and cancel before taking absolute values.

The exceptional character count and the first and second support payments in [1, (18.41)–(18.44)] are independent of κ . In the centered case their remaining deficit is at most

$$A - \frac{5}{6}M - \frac{2}{3}v - (L - v)_+ \leq (A - M)_+, \quad v = c + w + \min(c_2, d_2), \quad L = M/4.$$

Indeed its maximum as a function of $v \geq 0$ is attained at $v = L$ and equals $A - M$. For positive factors, admission implies $A \leq M$. For the padded zero-factor core, $A - M$ is at most its padding. This also covers exceptional rows admitted when a partition scalar or enclosing-ball weight is bounded by one after positivity. No exceptional row is silently supplied to (5).

3.4 Choice of mesh, height orders and induction losses

Prove the zero-factor width bands first and then the positive-factor bands, each in increasing width. Within a band prove the uncentered range $A \leq 5M/6$ before its comparison. Reflection gives arbitrary zero-factor lengths from the padded core; a comparison calls only the uncentered assertion already proved at that width.

Before specifying N , choose a width floor ρ , width step σ and padding δ_{pad} , with the latter small relative to the first two, such that

$$T_{\text{term}} := \rho + \delta_{\text{pad}} + \rho/6 + 3\sigma + 5\sigma/3 < \varepsilon/4.$$

These terms pay, respectively, the small-width estimate, its padding, the terminal prime cost, and the exceptional and diagonal losses. If the prescribed initial width is at most M_{max} , put $D = 2 + \lceil 2M_{\text{max}}/\sigma \rceil$. Let $C_* \geq 1$ bound the fixed number of aggregate affine perturbations in one stage, enlarged to include (9). Choose

$$\xi < \min \left\{ \frac{\delta_{\text{pad}}}{2}, \frac{\rho}{100C_*}, \frac{\sigma}{4C_*}, \frac{\varepsilon}{16C_*D} \right\}, \quad \eta < \min \left\{ \frac{\sigma}{6}, \frac{\varepsilon}{16C_*D} \right\}, \quad \varepsilon_0 < \frac{\varepsilon}{16C_*D}.$$

The affine admission changes by at most $|\Delta M| + |\Delta A| + 5|\Delta z|$, since $0 \leq 6\kappa - 1 \leq 5$. Apply this to aggregate subset errors. For the fixed slot system, the threshold on θ_N and endpoint logarithms, together with the aggregate tolerance, ensures $e_c \leq \rho/100$ and the comparison margins after (9). Every strict child then drops width by at least $\sigma/2$; there is no same-band cycle.

At the width floor the zero-factor core is estimated by absolute counting, and (6) with (5) pays its positive factors. Above the floor the second-transform diagonal costs at most $5\sigma/3$ plus the aggregate errors [1, (18.31)]. Together with the uncentered exceptional count, it first completes $A \leq 5M/6$ in the current band. Only then are the comparisons in Section 2 invoked. Exceptional raw scales are never supplied to a clipped moment assertion. These are precisely the terminal and same-band calls used in the induction.

For at most d remaining strict calls use the error envelope

$$\mathcal{E}_d = T_{\text{term}} + (d+1)C_*(\eta + \xi + \varepsilon_0).$$

The preceding bounds close backward induction with total error less than ε . All surviving subsets use the same mesh. Small divisor losses are subdivided after N is fixed; they are not charged as a fixed exponent once per prime or once per slot. Each discrete label mass is extracted once and its subsequent measure normalized before taking the smooth norm.

Finally fix the internal derivative and height orders backwards through this finite induction. Lemmas 4.5 and 4.7 of [1] bound the full weighted Fourier measures and all Euler derivatives of the radial kernels uniformly in their scale ratio. Increasing a derivative order increases a fixed seminorm and height degree, not a power $Z^{J\xi}$. The polynomial factor in (5) is integrated with these weighted measures. Only after these internal orders are fixed is the external Fourier-tail order chosen. Reflection, masks and all finite ray choices preserve the same bounds. This proves Theorem 1.1, including its original uniformity and mesh-before-factor-count assertion.

4 The witness count with the extended capacity

We apply Theorem 1.1 to the physical rows and witnesses of [1, Sections 8 and 19]. To distinguish two uses of q , write q_{amp} for the mean prime amplitude; the moving-twist length in this application is zero. Each row character has the fixed ray twist ν and the varying factor $\chi_n(u)$.

Proposition 4.1 (Actual witness count). *Suppose $13/18 \leq \kappa \leq 3/4$ and $\beta_* \leq (1 + \kappa)/2$. Fix a dynamic and amplitude bin of retained sixth-power-free physical rows $q_u \asymp U$, with*

$$1/50 < \delta = 2a - 1 \leq 2\beta_* - 1 \leq \kappa, \quad q_{\text{amp}} \in [0, \delta/2], \quad x = q_{\text{amp}}/\delta.$$

Assume the total available prime length at base U exceeds $7/37$ by a fixed positive amount, with disjoint underlying prime windows and the coefficients of [1, Section 19.2]. For $t \in [1, 3/2]$, put

$$c = \frac{1}{3\kappa}, \quad B_x = 2 - 2cx, \quad D_x = B_x + 1 - x, \quad P_x = B_x(1 - x), \quad \alpha = \frac{5}{6}.$$

Then, for every prescribed $\varepsilon > 0$, the strict capacity decrements and the prime mesh can be chosen so that

$$\#\mathcal{B} \ll U^{\max\{R_{\text{short}}(t), L(t)\} + \varepsilon} (1 + T_1)^{A_\varepsilon}, \quad (14)$$

where

$$R_{\text{short}}(t) = 1 - \delta + \frac{\delta P_x}{D_x}(3/2 - t), \quad L(t) = 1 - \delta + (\alpha - \delta)(t - 1).$$

The estimate uses the actual rowwise witnesses, profiles, cutoffs and height condition of [1, Proposition 8.3]. All moment, selection, rounding and parameter-recovery costs are included in ε and the fixed polynomial height factor.

Proof. Subdivide by the witness presentation and its dyadic length pair, as in [1, Proposition 19.2]. The logarithmic number of subdivisions is absorbed in the prescribed loss. For one subdivision write r, m for the inverse and plain witness lengths. The detector gives both inequalities

$$r + m \geq t - o(1), \quad r \leq t + o(1). \quad (15)$$

No equality between these lengths is assumed.

The inverse moment [1, Lemma 17.1] has capacity $z_I(r) = (1 - r)/2$ for $r < 1$. Applying Theorem 1.1 to two copies of the plain witness gives $z_P(m) = (1 - 2m)/(6\kappa)$ for $m < 1/2$. Order the positive prime amplitudes decreasingly and fill a requested capacity fractionally in that order. Its gain is at least $q_{\text{amp}}z$. Removing the last fractional factor costs at most δ times the largest mesh length in the squared spike. Approach each positive capacity from below by a fixed decrement. Both losses may be made arbitrarily small, with the mesh chosen before the number of physical prime factors.

The witness spikes and the two moments yield the count exponents

$$1 - \delta r - 2q_{\text{amp}}z_I(r), \quad 1 - 2\delta m - 2q_{\text{amp}}z_P(m).$$

The second denominator is the fourth power of the plain witness. It is not a second-moment estimate. Substitution, and then (15), give the decreasing and increasing affine functions

$$A_I(r) = 1 - \delta\{x + (1 - x)r\}, \quad S_t(r) = 1 - \delta\{cx + B_x(t - r)\}.$$

The replacement of m is legitimate because its exponent has derivative $-2\delta + 2\delta cx < 0$. They meet at

$$r_*(t) = \frac{B_x t + (c - 1)x}{D_x}. \quad (16)$$

Their common value is $R_{\text{short}}(t)$. Use the plain count for $r \leq r_*$ and the inverse count for $r \geq r_*$, except in the small zero-capacity neighborhoods.

In the whole range $4/9 \leq c \leq 6/13$, $0 \leq x \leq 1/2$, directly from (16),

$$r_*(1) \geq 23/37, \quad r_*(3/2) = 1, \quad \frac{1}{3} \leq t - r_*(t) \leq \frac{1}{2}.$$

Thus the requested inverse length is at most $7/37$. Its second strict condition is also retained: after decreasing z_I by $\nu_0 > 0$,

$$1 - r - 2z \geq 2\nu_0, \quad 3 - 2r - 8z = 4(1 - r - 2z) + (2r - 1) > 0.$$

For the inverse side, $r \geq 23/37$ makes the latter margin at least $9/37$ before its first positive term. On the plain side $m \geq 1/3 - o(1)$, so its requested length is at most $c/6 + o(1) \leq 1/13 + o(1)$. The stated supply margin therefore suffices for both selections.

For $r \geq 1$, use the unmarked sixth-power amplification of [1, Lemma 17.6, (17.88)]. Its normalized inverse energy has exponent $e(r) = \max\{1, (1 + 5r)/6\}$, with arbitrarily small uniform loss on bounded length ranges. Dividing by the witness spike gives $1 - \alpha + (\alpha - \delta)r$. Since $\delta \leq \kappa < \alpha$, the second inequality in (15) bounds this by $L(t) + o(1)$. For $m \geq 1/2$, the zero-factor fourth moment gives exponent $1 - 2\delta m \leq 1 - \delta$. This is below $R_{\text{short}}(t)$. If a positive capacity is too small for its decrement, use these corresponding unweighted bounds. Their uniform forms cover the shrinking neighborhoods of $r = 1$ and $m = 1/2$; no marked estimate is invoked at a vanishing strict margin.

It remains to verify the coefficient and parameter hypotheses. With $\Theta = \langle \eta, \hat{T} \rangle$, every physical prime coefficient is a fixed combination

$$\nu(p)\mathbf{1}_{p \in 1_{\mathcal{T}}} = |\mathcal{T}|^{-1} \sum_{\vartheta \in \hat{\mathcal{T}}} (\nu\vartheta)(p), \quad \nu \in \Theta.$$

If necessary, conjugate the *whole* witness and selected-prime product; conjugate profiles and zeros with it. This gives the orientation $\nu(n)\chi_n(u)$ of Theorem 1.1 without altering an underlying prime support. A retained large sixth-power-free row has a good prime of valuation $1, \dots, 5$, so its inducing character ramifies there and lies outside Θ . Rows supported entirely on $\mathcal{S}$, including all unit rows, form the fixed finite set handled by the separate small-row argument.

The selected real parts lie in a bounded smooth family. Recover the rowwise selected real parts, heights and scales using the common parameter-Sobolev measure of [1, Lemma 4.5 and Section 19.2]. Derivatives insert logarithmic profile weights, and the height range contributes the fixed power of $1 + T_1$. These selections do not authorize row-dependent prime coefficients. Fix all internal moment orders before the external recovery-tail order. This retains the original cutoffs, natural zeros, independent prime windows and cumulative frequency losses, proving (14). $\square$

For reference, the balance of the two branches is explicit. Put

$$\mathcal{J} = (\alpha - \delta)D_x + \delta P_x > 0.$$

Then $D_x(R_{\text{short}} - L) = \delta P_x/2 - \mathcal{J}(t - 1)$, so

$$t_* = 1 + \frac{\delta P_x}{2\mathcal{J}}, \quad R_* = 1 - \delta + \frac{(\alpha - \delta)\delta P_x}{2\mathcal{J}}.$$

No replacement by a fixed baseline capacity, and consequently no additional baseline-capacity loss, is needed. The endpoint certificate, Euler factors, residues and continuation using this count are separate from the present supplement.

The remaining elementary branches are unchanged. With no selected primes, take $t = 1$; the unweighted inverse and fourth moments give count exponent $1 - 2\delta/3$ by balancing $r = 2/3$, $m = 1/3$. The detector's floor bin $a = 51/100$ uses the original trivial row count, of exponent one, rather than a positive-slot estimate. The fixed finite small-row set is retained separately as described above. None of these branches needs a new admission or coefficient.

References

- [1] OpenAI, *The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane $\Re s > 7/8$* , unrefereed preprint, 30 September 2026. All numbered references in this supplement use commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. [Pinned manuscript](#). The corresponding TeX source has SHA256 `42a5ee0febca59fd1def55cfd6c6808c322ef4237d7726303ea52f711deac6a3`.